

At times, there were calls for protests seeded in the same location at the same time, for groups with opposing views on an issue. The Russian campaign demonstrated the nefarious uses to which Facebook ads could be put to, for a relatively small amount of cash. All the ads purchased by the entities linked to Russia totalled about \$100,000. The total amount of ad spend by the Clinton and Trump campaigns was a staggering \$81,000,000.

A single company provided social media election campaign support to three Republican nominees in the 2016 US Presidential election, Ted Cruz, Ben Carson and Donald Trump. The approach by the company involves something that is known as psychographic profiling. The profiles understand the value systems of an individual, their motivations, their fears, their attitudes, as well as their behavior. At only 70 likes, the company can predict personal attributes such as skin colour or sexual orientation. Around 150 likes are enough to understand an individual better than their parents. More than 350 likes, and the company knows the person better than they know themselves. Autonomous systems work out the profiles down to an individual level, and then serve them fine tuned ads that are likely to push the targets to perform a desirable action. There are reports that the UPA is in touch with Cambridge Analytica to fine tune the strategy for future elections. The Economic Times referred to the services of the company as the "Brahmastra".

Facebook ads can potentially be used for surveillance as well. University of Washington researchers have demonstrated the use of Facebook ads for tracking a person. The ads are targeted to a device id, known as Mobile Advertising ID, as well as geolocation triggers that are spaced out in a grid across a city. As and when the user visits a particular location, the ad is delivered to the person. The malicious actors who purchased the ads can track the movements on the individual by checking where and when the ad is getting delivered. An individual can execute such an attack with a modest budget, by purchasing less than \$1,000 worth of Facebook ads. ■

## HARDWARE AND SOFTWARE VULNERABILITIES

There are fundamental flaws in hardware and software that repeatedly allow attackers to breach systems. The Meltdown and Spectre CPU bugs affect virtually all modern devices, and are independent of operating systems. Although no known attack has been carried out through these bugs, the attackers can potentially retrieve sensitive information such as usernames and passwords using malicious code that is executed through a tab in a browser window. Fixing these vulnerabilities for the end user is not easy, as it involves updating the BIOS, and requires specific patches from the manufacturers of the motherboards. Not all manufacturers have issued patches for all their motherboards, which leaves the older systems particularly vulnerable. In some cases, the security patches are likely to slow down the computer. When the fixes were released for Linux systems, the developers actually scrubbed the code of all comments, so that it is harder for cybercriminals to understand and exploit the problem. Considering that the flaw is related to a fundamental design of the chips, Meltdown and Spectre will continue to be a cybersecurity concern for the foreseeable future. For now, it is safest to have only one browser tab open when conducting financial transactions, and to start using the internet normally after having closed the window used for the transaction for some time.

According to Trend Micro, systems will continue to be compromised in 2018 through known vulnerabilities, for which patches have already been issued. Two problems that are expected to aggravate the issue is that a number of IoT devices will be manufactured without adequate industry standards, as well as an increase in connectivity. Of course, new vulnerabilities will also be exposed.

The problem here is that entities that become aware of these vulnerabilities do not always disclose them to the technology companies involved. The NSA developed tools to exploit these vulnerabilities, which were subsequently used to execute attacks against the USA. After Edward Snowden blew the whistle on the NSA spying campaign, lawmakers made the NSA agree to disclose critical vulnerabilities such as HeartBleed, instead of hoarding them. The CIA however, spun out its own spying campaign, and these tools are documented in the Vault 7 disclosures by WikiLeaks.

Although Google is pretty good at both researching vulnerabilities as well as patching them for the Android platform, the smartphone vendors may not be as quick to roll out the updates to users using variants of the OS. This leaves hundreds of thousands of Android users exposed to attacks. The most dangerous kinds of malware that could infect Android phones include ransomware, cryptocurrency miners and applications that conduct transactions with the knowledge of the user.

In 2017, a crippling flaw was exposed in a code library developed by Infineon, a German chipmaker. The library was used to generate encryption keys for everything from national ID schemes to government computers. The generation of the easy to hack keys has been going on since 2012. Kaspersky predicts more fundamental flaws in cryptographic techniques will emerge in 2018.

Users can minimise their exposure to malware that exploit vulnerabilities by continuously keeping their devices and machines updated with the latest security patches. If the update log shows only bug fixes and security patches, it is not a good idea to avoid installing that update. ■